PROGRAM -6

Static and Shared libraries

Write a program to create and use static and shared libraries. Demonstrate the advantage of shared libraries over static libraries in terms of memory usage.

To generate a static library (object code archive file):

- Compile: `cc -Wall -c ctest1.c ctest2.c`

- Create library "libctest.a" by archiving it:

```
ar -cvq libctest.a ctest1.o ctest2.o
```

- List files in library:

```
ar -t libctest.a
```

- Linking with the library:
  - Library in current directory:

```
cc -o executable-name prog.c libctest.a
```

  - Library in not in current directory:

```
cc -o executable-name prog.c
-L/path/to/library-directory -lctest
```

- `size executable-name`

To generate a shared object (Dynamically linked object library file):

- Compile: `gcc -Wall -fPIC -c *.c`
  Here PIC meaning generate position independent code
- Create shared library "libctest.so"
  `gcc -shared  -o libctest.so   *.o`

- export LD_LIBRARY_PATH=.:LD_LIBRARY_PATH
- Linking with the library:

```
gcc  -L . prog.c -l ctest -o prog
```

- `size prog`

Compare the size of shared and static libraries using size command